

SILTHAUL · REQUIREMENTS

SiltHaul requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-03	Amish Chadha	Amish's requirement decisions (SLH-DDR-003); R1 restated (no lifting or carrying of mud; at least 0.5 times the bucket crew) and met on paper with a second box on the return leg; R9 met on paper with a flat-pack frame; status rerun from SLH-CAL-001 v0.3
0.3	2026-10-03	Amish Chadha	TRL 3 status from SLH-CAL-001 on the constructable design (SLH-DDR-002); R10 restated against the value-engineering target
0.2	2026-10-03	Amish Chadha	TRL 2 requirements with concept status; R11 overload limit and R12 load holding added (SLH-DDR-001)
0.1	2026-09-30	Amish Chadha	Initial scaffold

SiltHaul requirements

Eleven of the twelve requirements are met on paper or by design, one (R8, setup time, now estimated at the 20 min limit) can only be verified by a timed trial, and none is reported not met (SLH-CAL-001 v0.3, Table 3). On 2026-10-03 Amish chose option A on every requirement decision put to him ("1A 2A 3A 4A 5A 6A 7A 8A 9A 10A 11A"), which for SiltHaul meant two changes (SLH-DDR-003). Under 2A, R1 is restated as "no lifting or carrying of mud; output at least 0.5 times the same crew with shovels and buckets", and a second box rides on the return leg so the rope loop carries mud out on both strokes: about 0.51 m³ an hour with rest, a ratio of 0.57. Under 7A, the frame is a bolted flat-pack frame so the kit fits a small car boot, with the 25 kg heaviest-part limit kept. R11 and R12 were added at TRL 2 under Amish's pre-approval of 2026-10-03 (SLH-DDR-001), because the anchors and rope can only be sized on a known maximum rope tension, and a hand-cranked load must hold when the handles are let go. "Met on paper" means shown by calculation, not by test.

Safety: SiltHaul is a rope-and-anchor system under tension with a hand-cranked chain drive. R6, R7, R11 and R12 are its safety requirements; they are checked on paper only and must be proved by proof-load tests before anyone uses the kit (SLH-BLD-001, section 6).

Table 1. Requirements and status at TRL 3

ID	Requirement	Target	Verification (TRL 3 or later)	Status (SLH-CAL-001)
R1	Mud moved per hour by a crew of four	No lifting or carrying of mud; output at least 0.5 times that of the same crew with shovels and buckets (estimate)	Side-by-side timed trial on a test bed of wet silt	Met on paper: 0.51 against 0.90 m ³ /h (ratio 0.57); loaders push mud in over the lip; nobody lifts or carries it
R2	Crank handle force at full box	150 N (34 lbf) or less per person with two people at the cranks	Spring balance at the handle during trial	Met on paper: 63 N each at 1,000 N (126 N for one person); 61 N each at the 970 N stroke estimate
R3	Scraper box capacity	About 40 L (10 US gal) of wet mud per trip	Measured fill on test bed	Met by design: 40.3 L in each of the two boxes at a 245 mm fill
R4	Fits through a doorway	Box, tail block and ramp pass a 0.7 m (28 in) opening, with the return leg beside the box	Fit test	Met by design: each box 258 mm over the shackles; the boxes pass 42 mm apart and cross the ramp 26 mm inside its cheeks; ramp 646 mm
R5	Working length	Up to 15 m (49 ft) from tail sheave to capstan	Trial at full length	Met by design: 12 m of travel on each drum half; capstan at least 6 m from the door
R6	Rope and anchor strength	Minimum break strength at least 5 times the working pull, and at least 5 times the R11 limit for the rope and chain	CalRig proof-load test of each anchor type	Met on paper: rope 18 times working, 5.4 at the limit; chain 5.0
R7	Tail anchor independent of walls	Holds the R11 limit on both legs on a bare concrete floor with no wall contact	Pull test on the anchored tail block	Met on paper on a sound slab: 0.45 of recommended anchor loads
R8	Setup time	Three people set up in 20 min or less	Timed trial	Not verifiable at TRL 3: about 20 min estimated, at the limit (bolting the frame and shackling two boxes add 4 min)
R9	Portability	Complete kit fits in a car boot; heaviest part 25 kg (55 lb) or less	Weigh and pack test	Met on paper with the rear seats folded: heaviest lift 23.1 kg (drum with bearings); flat-pack frame, largest piece 890 x 814 x 83 mm; kit about 454 L packed against a 594 L small car boot
R10	Cost	Prototype parts at or under the USD 2,000 value-engineering target (a hypothetical control target)	Costed bill of materials	Met on paper, within the target: USD 1,055.60 (USD 944.40 under)
R11	Overload limit	Rope tension limited to 3,000 N whatever force the crew applies	Shear pin release test on the capstan	Met on paper: shear pin releases at 2,451 N (1,961 to 2,941 N)
R12	Holds when let go	The drum holds the load in either direction when the cranks are released	Release test at working pull	Met by design: two ratchet wheels and pawls on the drum shaft

■ Requirements at risk or not met

None is reported not met. Two results are close to their limits and are watched at TRL 4:

- **R1.** The ratio of 0.57 rests on estimated loading, dumping and bucket rates; a timed trial settles it. The restatement is Amish's (2A, 2026-10-03): "no lifting or carrying of mud; output at least 0.5 times the same crew with shovels and buckets".
- **R8.** About 20 min with three people, at the limit, now that the frame is bolted together on site and two boxes are shackled.
- **R9.** Met with the rear seats of a small hatchback folded. With the seats up the 890 x 814 mm side frames do not lie flat in a typical small boot; see the design decisions register.

■ Assumptions

- Most flood mud is fluid enough to be scraped and dragged, not dug; very stiff or deep deposits may still need shovels first.
- The building has a sound concrete slab at least 100 mm thick for the tail block anchors; where it does not, SiltHaul is not used in that room.
- Volunteers accept a short briefing on rope safety.
- A small car boot is taken as a small hatchback's load space with the rear seats folded: 1,250 mm long, 950 mm wide and 500 mm high (about 594 L).
- Loaders push or rake mud into the box over its lip at floor level; the box is 305 mm tall, so shovelling over the side is possible but not needed.
- A safe dump point exists outside within rope reach, and an anchor point (a sound tree or a parked vehicle) exists behind the capstan.